

Nithin Rajan

Robotics Engineer with extensive experience in developing and implementing cutting-edge robotic systems and automation solutions. Skilled in C++, Python, ROS 1/2, and various sensor technologies, with Practical expertise in Simulation, Sensor Calibration, and Real-time data processing. Proven ability in system automation testing, Embedded systems, and Self-driving vehicle technologies. Strong leadership in team management, algorithm optimization, and delivering high-quality project outcomes. Successfully executed complex projects across the Automotive, Infotainment, and educational robotics sectors.



Work History

2022-07 -
2025-08

AUTOMATION AND VALIDATION SPECIALIST,

TATA ELXSI, Trivandrum, India

Technical Tasks

- Requirement Analysis of System & Software Requirements, and Develop the Testcases.
- Test Bench Preparation for the Validation, and resolve there when there is any issue.
- Analyse the Feasibility of the Automation and Manual Execution Testcases.
- Preparation of the CANoe Configurations as per the Software Release plan updates.
- Test Execution Support, Review of the Bugs and Bug Creation in the Redmine Platform.
- Designed and executed Hardware In Loop (HIL) testing using VT cards to simulate the OHC board module and capture critical data values.
- Utilized CAPL scripting to automate test cases for features and diagnostics in CANoe 16 and VTest Studio.
- Developed diagnostic and feature scripts for CAN and Ethernet communication using Python programming and BDD framework.
- Discussion with the Higher Management and Customers for the System Testing. Developed comprehensive validation plans, outlining objectives, timelines, and resource requirements for successful project execution.

Contact

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LinkedIn

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Skills

- Python ●●●○○
- C ●●●●○
- C++ ●●●●○

- CAPL Scripting ●●●●○
- CANoe ●●●○○
- VTest Studio ●●●●○
- ROS1 / ROS2 ●●●●○
- Mujoco ●●○○○
- Gazebo ●●●○○
- Issac Sim ●●○○○
- JIRA ●●○○○
- Git proficiency ●●●○○
- Linux ●●○○○
- Arduino Programming ●●●○○
- UDS ●●○○○
- CAN ●●●○○
- LIN ●●●○○
- PyTorch ●●○○○

Personal Details

Date of Birth: 19/07/1989

Nationality: Indian

Marital Status: Married

Visa Status: Resident Visa

2019-05 -
2021-09

- Provided support and training to the ADAS interns for Internals projects involving C++ implementation of ROS1/ROS2 in Localization & Navigation, Robot Arm Manipulation (Move IT), Image Processing, and Sensor Calibration & Simulation (Gazebo).

Management Tasks:

- Creating the Review Reports for the Following:
 - Review the Report preparation of the Testcase Development Documents, after the changes with respect to System and Software Requirements.
 - Peer Review Report preparation for the Test Report before delivery to the customer.
 - Peer Review Report preparation of the CANoe Configuration before delivery to the customer.
 - Review the CAPL on the CANoe Configuration and the CAPL Scripts.
- Capturing into the Document the Coverage of Requirements and the Testcases covering the Requirements.
- Maintain the JIRA Tickets for Plan, Monitor, and Track the day-to-day tasks of the Team Members.
- Created comprehensive test plans to guide test execution
- Create the Traceability Report from Requirement to Testcase Development Document, Report to Testcase Development to Document, and Compliance Matrix to Testcase Selection using the Rectify Tool.

SENIOR SOFTWARE ENGINEER

DEEVIA SOFTWARE INDIA PVT LTD, Bengaluru, India

- Ported C implementation to PP (Pixel Processing) implementation for the Trailer Tracking Test.

Languages

English	
	Upper intermediate (B2)
Malayalam	
	Bilingual or Proficient (C2)
Hindi	
	Intermediate (B1)
Tamil	
	Intermediate (B1)

- Optimized and tested the Bilinear Interpolation Algorithm in the PP implementation for the Trailer Tracking Test.
- Developed resolutions for different camera views as specified by the Customer.
- Created and tested golden images for various camera views based on updated software requirements from the BMW team (BEGU generation offline tool).
- Integrated and tested software components on the embedded platform.
- Created project release packages, uploaded code for Gerrit review in GITHUB, and maintained the project repository.
- Monitored, updated, and resolved bugs reported in JIRA tickets..
- Delivered unit-tested systems within customer-prescribed timeframes.
- Collaborated with fellow engineers to evaluate software and hardware interfaces.

2018-09 -
2019-05

SOFTWARE ENGINEER

- DEEVIA SOFTWARE INDIA PVT LTD, Bengaluru, India
- Developed a ROS1 MoveIt model for the Panda Arm robot, incorporating a custom-made vacuum gripper.
 - Optimized trajectory planning within the OMPL interface.
 - Integrated the OMPL interface for a new planner.
 - Developed the C++ implementation of datalogging for different parameters across various modules for the Rear Camera project

2017-12 -
2018-09

ROBOTICS ENGINEER

- ROBOWARE TECHNOLOGIES LLP, Bengaluru, India
- Developed, tested, and optimized existing algorithms using ROS for the Wipod, executed on the Arduino embedded platform.
 - Researched, implemented, and tested the Navigation, Localization, and 3D mapping stack in ROS.

- Delivered demonstrations of the L-3 Wipod vehicle model to customers.
- Programmed Arduino for vehicle movement and LIDAR actuation.
- Tested and calibrated LIDAR and Laser sensors for the Navigation stack in ROS.

2015-06 -
2016-12

ROBOTICS ENGINEER

NOVATECH ROBO PVT LIMITED, Bengaluru, India

- Delivered comprehensive training sessions on fundamentals of electronics, embedded systems (8051), and Arduino programming, along with simulation tools like Proteus8 and MATLAB.
- Organized and facilitated interactive workshops featuring platforms such as Novabot, LEGO Mindstorms, 8051 development boards, Arduino, and MATLAB, enhancing technical skills among students and peers.
- Engineered a variety of robotic solutions—including autonomous mobile robots, robotic arms, and drones—for real-world academic applications.
- Improved robotic system efficiency by designing and implementing hardware upgrades.
- Utilized simulation software to validate proposed solutions before physical implementation, reducing development time and costs.
- Built, configured, or tested robots or robotic applications.
- Tested features, documenting findings, modeled trends, and submitted reports.

Education

2015-06

Master of Technology: Robotics

SRM University - Chennai, India

- Completed the post-graduation from SRM University, Chennai with an 8.732/10 CGPA

2011-07

Bachelor of Technology: Electrical And Electronics Engineering

Calicut University - Thrissur, India

- Completed the Under Graduation from Calicut University with 61% in the aggregate.

Certifications

- Certifications in ROS1 / ROS2 [Navigation, Manipulation, Localization, and Sensor Fusion]- Udemy
- Certifications in C++ - Udemy
- Certifications in Python - Udemy

Declaration

I hereby declare that the above-written particulars are true to the best of my knowledge and belief.

NITHIN RAJAN